

Network Testing Tools

Task – 1 : Use the ping command to test connectivity from your computer to www.youtube.com and note the average response time.

Solution : Ping

```
PS C:\Users\Vaidehi> ping youtube.com

Pinging youtube.com [142.250.143.190] with 32 bytes of data:
Reply from 142.250.143.190: bytes=32 time=28ms TTL=110
Reply from 142.250.143.190: bytes=32 time=68ms TTL=110
Reply from 142.250.143.190: bytes=32 time=42ms TTL=110
Reply from 142.250.143.190: bytes=32 time=37ms TTL=110

Ping statistics for 142.250.143.190:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 28ms, Maximum = 68ms, Average = 43ms
PS C:\Users\Vaidehi>
```

Task – 2 : Run the traceroute (tracert on Windows) command to www.flipkart.com and list all the intermediate hops shown in the output. Hint : Each hop represents a network device your data passes through to reach the destination.

Solution : Traceroute to Flipkart

```
PS C:\Users\Vaidehi> tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.115]
over a maximum of 30 hops:

  1  74 ms    3 ms     4 ms   10.180.115.166
  2  715 ms   17 ms    27 ms   192.168.76.130
  3  17 ms    18 ms    11 ms   192.168.116.132
  4  41 ms    49 ms    22 ms   192.168.116.177
  5  49 ms    27 ms    23 ms   192.168.251.91
  6  47 ms    13 ms    26 ms   125.21.143.30
  7  87 ms    85 ms    47 ms   125.21.143.29
  8  50 ms    51 ms    34 ms   116.119.36.180
  9  57 ms   143 ms   41 ms   ae8.gpx-bom2.netarch.akamai.com [23.55.247.248]
 10  47 ms    79 ms    37 ms   a23-55-244-115.deploy.static.akamaitechnologies.com [23.55.244.115]

Trace complete.
PS C:\Users\Vaidehi>
```

Task – 3 : Use the nslookup tool to find the IP address of www.zomato.com and record the result.

Solution : Find Zomato's IP Address Using

```
PS C:\Users\Vaidehi> nslookup zomata.com
Server: UnKnown
Address: 10.180.115.166

Non-authoritative answer:
Name: zomata.com
Address: 185.80.7.13
```

```
PS C:\Users\Vaidehi> |
```

Task – 4 : Test connectivity to another device on your local network (such as your friend's laptop or your phone) using the ping command and describe what happens if the device is offline.

Solution : Ping a Device on Your Local Network

```
Default Gateway . . . . . :  
  
Wireless LAN adapter Wi-Fi:  
  
    Connection-specific DNS Suffix  . :  
    IPv4 Address. . . . . : 10.180.115.246  
    Subnet Mask . . . . . : 255.255.255.0  
    Default Gateway . . . . . : 10.180.115.166  
PS C:\Users\Vaidehi> 10.180.115.247  
PS C:\Users\Vaidehi> ping 10.180.115.247  
  
Pinging 10.180.115.247 with 32 bytes of data:  
Reply from 10.180.115.246: Destination host unreachable.  
Reply from 10.180.115.246: Destination host unreachable.  
Reply from 10.180.115.246: Destination host unreachable.  
Reply from 10.180.115.246: Destination host unreachable.  
  
Ping statistics for 10.180.115.247:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
PS C:\Users\Vaidehi>
```

Task – 5 : Create a short report (5-7 lines) summarizing which tool (ping, traceroute, nslookup) is most useful for diagnosing each of these problems: (a) Website not loading, (b) Slow connection, (c) Suspecting DNS issues. Constraint : Give a specific example for each tool based on your own test results above.

Solution : Which Tool Is Useful for Which Problem?

The **ping** command is useful for checking whether a website or device is reachable and measuring response time. In my test, I used ping www.youtube.com and received an average response time of ___ **ms** with ___% **packet loss**. The **tracert** command is useful for identifying the path and intermediate network hops between my computer and a destination. I used tracert www.flipkart.com and observed ___ **hops** in the output. The **nslookup** command is useful for diagnosing DNS problems because it shows the IP address associated with a domain name. I used nslookup www.zomato.com and obtained the IP address ___. Therefore, ping is useful for connectivity and delay, tracert for path-related problems, and nslookup for DNS-related issues.